

A non-Gaussian limit for nodal surplus

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1 Statement

On a finite connected metric graph with at least one edge and no degree-two vertices, consider $-f''$ with continuity and Kirchhoff conditions at every vertex. Enumerate eigenvalues with multiplicity, starting with $\lambda_1 = 0$. An index is *generic* if its eigenvalue is simple and its eigenfunction is nonzero at every vertex. Its nodal surplus is the number of interior-edge zeros minus $n - 1$. For positive rationally independent lengths, the frequencies among generic indices define a probability law S , supported on $\{0, \dots, \beta\}$ and symmetric about $\beta/2$, where $\beta = |E| - |V| + 1$ [1, Theorem 2.1]. The spectral-frequency limit is always taken before the graph-size limit.

For $m \geq 3$, take vertices $u, v, w_1, \dots, w_m, t$. Each module has two parallel edges from u to w_i and one edge from w_i to v ; add the pendant edge ut . The graph is loop-free, has no degree-two vertices, and has $\beta_m = 2m - 1$. Number the $3m$ core edges module by module. With p_j the j th prime, assign lengths

$$\ell_j = \sqrt{p_j} \quad (1 \leq j \leq 3m), \quad \ell_0 = m^6 \sqrt{p_{3m+1}} \quad \text{on } ut. \quad (1)$$

They are rationally independent: in the multiquadratic field generated by the prime square roots, changing one root's sign isolates its coefficient in any rational relation. Let S_m be the resulting spectral surplus variable.

Here is an explicit description of the limit. Let X_1, X_2 be independent standard Cauchy variables and ϵ_1, ϵ_2 independent fair signs, independent of them. Put

$$W = \frac{\epsilon_1 \sqrt{1 + X_1^2} + \epsilon_2 \sqrt{1 + X_2^2}}{\sqrt{1 + (X_1 + X_2)^2}}, \quad \mathbb{E} e^{i(tA+sB)} = \exp\{-\mathbb{E}|tW + s|\}. \quad (2)$$

Existence of (A, B) is proved below. For three fresh independent standard Cauchy variables, set $C_j = \sqrt{1 + \widehat{X}_j^2}$ and define

$$v(r) = \frac{1}{2} - \frac{1}{4} \mathbb{P}(|C_1 - C_2| < |r| C_3 < C_1 + C_2), \quad R = -A/B, \quad V = v(R), \quad a_* = \mathbb{E}V. \quad (3)$$

The probability defines a deterministic function v ; its auxiliary variables need not be coupled to W .

Theorem 1. *The variable R is finite and nonzero almost surely, $1/4 < V < 1/2$ almost surely, and $\text{Var}(V) > 0$. For $Z \sim N(0, 1)$ independent of V , writing $X_m = (S_m - \beta_m/2)/\sqrt{m}$, one has*

$$\mathbb{E}S_m = \beta_m/2, \quad X_m \implies Z\sqrt{V}, \quad \mathbb{E}X_m^2 \longrightarrow a_*, \quad \mathbb{E}X_m^4 \longrightarrow 3\mathbb{E}V^2. \quad (4)$$

Consequently,

$$\begin{aligned} \frac{\text{Var}(S_m)}{\beta_m} &\longrightarrow \frac{a_*}{2} \in (1/8, 1/4), & \frac{S_m - \beta_m/2}{\sqrt{\text{Var}(S_m)}} &\implies Z\sqrt{V/a_*}, \\ \frac{\mathbb{E}(S_m - \beta_m/2)^4}{\text{Var}(S_m)^2} &\longrightarrow 3 \frac{\mathbb{E}V^2}{a_*^2} > 3. \end{aligned} \quad (5)$$

The standardized limit is not Gaussian. Thus this sequence contradicts the unrestricted Gaussian assertion in [2, Conjecture 3.1], not its separate universal linear-variance assertion.

2 Exact reduction to independent modules

We use one published spectral-measure fact. For a loop-free standard graph there are length-independent edge measures ν_e on generic secular phases. Their surplus laws $\mathcal{W}_e$ are symmetric about $\beta/2$, and

$$\mathcal{L}(S) = \sum_e \frac{\ell_e}{\sum_f \ell_f} \mathcal{W}_e, \quad (\pi_e)_\# \nu_e = \text{uniform measure on } [0, 2\pi)^{E-1}, \quad (6)$$

where π_e drops phase e [2, Proposition 6.1, Eq. (6.7), Lemma 6.7]. The measures are restricted to the generic set and normalized there.

Let $\tilde{S}_m$ have the pendant-edge law $\mathcal{W}_0$. Under this measure the core phases θ_{ij} are independent uniforms on $[0, 2\pi)$. For each module put

$$\begin{aligned} x_{ij} &= \cot \theta_{ij}, & c_{ij} &= \csc \theta_{ij}, & a_i &= x_{i1} + x_{i2} + x_{i3}, & p_i &= c_{i1} + c_{i2}, & q_i &= c_{i3}, \\ \xi_i &= \left(\frac{p_i q_i}{a_i}, -x_{i3} + \frac{q_i^2}{a_i} \right), & (U_m, T_m) &= \sum_i \xi_i, & R_m &= -U_m/T_m. \end{aligned} \quad (7)$$

Define the *centered* module function

$$h_i(r) = \frac{\operatorname{sgn} a_i}{2} [1 - \operatorname{sgn}(p_i + r q_i)(\operatorname{sgn} c_{i1} + \operatorname{sgn} c_{i2} + \operatorname{sgn}(r q_i))], \quad r \neq 0. \quad (8)$$

Choose right-continuous values as a function of $|r|$ on each half-line at its possible jump. Fixed denominator-zero sets are null and can be assigned arbitrary values.

Lemma 1 (Exact surplus formula). *On the independent-core-phase space,*

$$\tilde{S}_m - \frac{2m-1}{2} = \sum_{i=1}^m h_i(R_m) - \frac{1}{2} \operatorname{sgn} T_m. \quad (9)$$

Proof. For an edge ab of phase θ , the outgoing vertex matrix adds $\begin{pmatrix} -\cot \theta & \csc \theta \\ \csc \theta & -\cot \theta \end{pmatrix}$ on its two endpoints. Call the assembled matrix M . At a generic eigenvalue $k^2 = \lambda_n > 0$ with vertex vector F and all edge sines nonzero, one has $MF = 0$. Put

$$D = \sum_e \lfloor k \ell_e / \pi \rfloor, \quad Q = \sum_{e=ab} \mathbf{1}_{\{F_a F_b \csc(k \ell_e) < 0\}}.$$

On each edge, a sine wave has $\lfloor k \ell_e / \pi \rfloor$ zeros and one extra exactly when the indicated endpoint sign is negative. Thus the zero count is $D + Q$. Split the quadratic form $\int (|f'|^2 - k^2 |f|^2)$ into vertex-zero functions and edgewise Helmholtz extensions of vertex values. Integration by parts makes the two spaces orthogonal, with negative indices D and $n_+(M)$, respectively; the extension form is $-k F^\top M F$, and n_+ counts positive eigenvalues. Hence $n - 1 = D + n_+(M)$ and

$$s = Q - n_+(M) = Q - n_+(M_{\setminus u}) \quad (F_u \neq 0). \quad (10)$$

The last equality follows by using the kernel vector F in place of the u th coordinate vector: M is congruent to $0 \oplus M_{\setminus u}$. Formula (10) is phase-periodic; at a secular point, take positive phase lifts as lengths and $k = 1$.

Normalize $F_u = 1$. Eliminating w_i gives $F_{w_i} = (p_i + r q_i)/a_i$, and the equation at v is $U_m + T_m r = 0$. For completeness these operations, and generic completion, hold almost everywhere. Write

$$\Lambda_m = \sum_i (-x_{i1} - x_{i2} + p_i^2/a_i), \quad g_m = \Lambda_m - U_m^2/T_m.$$

Every exceptional equation involving $\sin \theta_{ij}, a_i, U_m, T_m, p_i + R_m q_i, g_m$ has, after clearing denominators, a nonzero trigonometric-polynomial numerator. Indeed, at $\theta_{ij} = \pi/4$,

$$a_i = 3, \quad U_m = 4m/3, \quad T_m = -m/3, \quad R_m = 4, \quad F_{w_i} = 2\sqrt{2}, \quad g_m = 6m.$$

Their zero sets are therefore null. Off them the pendant equation is $\tan \theta_0 = -g_m$, with exactly two solutions modulo 2π . Both have nonzero sine and cosine, and $F_t = 1/\cos \theta_0$. The pivots of $M_{\setminus u}$ are $-a_1, \dots, -a_m, T_m, -\cot \theta_0$, all nonzero. Thus both completions have a one-dimensional kernel and nonzero vertex values. Since every edge sine is nonzero, interpolation identifies this kernel with the eigenspace. No completion with $F_u = 0$ exists: the core matrix with u deleted is invertible, forcing the core to vanish, and the pendant would then have both value and derivative zero at u .

In (10) the pendant sign count and its positive pivot both equal $\mathbf{1}_{\{\tan \theta_0 < 0\}}$ and cancel. The three sign counts in module i , minus $\mathbf{1}_{\{a_i < 0\}}$, equal $1 + h_i(r)$ by (8). The remaining pivot contributes $\mathbf{1}_{\{T_m > 0\}}$. This proves (9) for both completions. Uniform projection in (6) now gives the asserted law; no conditional weighting of the two completions is needed. $\square$

3 One module and its boundary response

Suppress the module index. Set $\sigma = \operatorname{sgn} a$ and write $h(r) = \sigma g(r)$ using (8). The function g depends only on (c_1, c_2, c_3) . If their effective signs $(c_1, c_2, r c_3)$ agree, then $g = -1$; otherwise $g \in \{0, 1\}$. Thus $|h(r)| \leq 1$. The measure-preserving involution $\theta_j \mapsto \pi - \theta_j$ fixes every c_j and negates a . Consequently σ is a fair sign independent of (c_1, c_2, c_3) , and

$$\mathbb{E}h(r) = 0. \tag{11}$$

A uniform phase has $(\cot \theta, \csc \theta) \stackrel{d}{=} (X, \epsilon \sqrt{1 + X^2})$, with X standard Cauchy and ϵ an independent fair sign. Condition on $C_j = |c_j|$ and average the signs in (8). With $\rho = |r|$, the four cases obtained by fixing the third effective sign positive give

$$\mathbb{E}[h(r)^2 \mid C_1, C_2, C_3] = \frac{1}{4} \left(1 + \mathbf{1}_{\{\rho C_3 > C_1 + C_2\}} + \mathbf{1}_{\{\rho C_3 < |C_1 - C_2|\}} \right).$$

Therefore $\mathbb{E}h(r)^2 = v(r)$. The law of each C_j is continuous with support $[1, \infty)$. For every finite $r \neq 0$, both the triangle event in (3) and its complement contain nonempty open sets, so

$$\frac{1}{4} < v(r) < \frac{1}{2}, \quad v \text{ is even and continuous,} \quad \lim_{r \rightarrow 0} v(r) = \lim_{|r| \rightarrow \infty} v(r) = \frac{1}{2}. \tag{12}$$

Continuity follows by conditioning on C_1, C_2 to exclude boundary equalities, and then dominated convergence; the two limits follow the same way. In particular v is nonconstant.

Lemma 2 (Stable boundary ratio). *The response vector in (7) satisfies*

$$\xi \stackrel{d}{=} (W \sqrt{1 + X^2}, X), \quad W \text{ independent of } X, \quad \mathbb{E}W^2 = 2, \quad \mathbb{E}\|\xi\|^{3/4} < \infty. \tag{13}$$

Furthermore $(U_m/m, T_m/m) \implies (A, B)$ with the law (2). This limit has a continuous density and full support on $\mathbb{R}^2$. Consequently $R_m \implies R$, the ratio R has no atoms and full support on $\mathbb{R}$, and $\operatorname{Var}(v(R)) > 0$.

Proof. Put $s_0 = x_1 + x_2$ and $\varphi = \theta_3 - \arctan s_0$. Direct substitution gives

$$\xi^{(2)} = \tan \varphi, \quad \xi^{(1)} = \frac{p}{\sqrt{1 + s_0^2}} \sec \varphi.$$

Conditionally on the first two phases, φ is uniform. Its tangent is standard Cauchy and the sign of its cosine is an independent fair sign. Since $p/\sqrt{1 + s_0^2}$ has the symmetric law of W , this proves the response representation. Averaging the signs in W and putting $z = x + y$ gives

$$\begin{aligned} \mathbb{E}W^2 &= \frac{1}{\pi^2} \int_{\mathbb{R}^2} \frac{2 + x^2 + y^2}{(1 + (x + y)^2)(1 + x^2)(1 + y^2)} dx dy \\ &= \frac{1}{\pi^2} \int_{\mathbb{R}} \frac{dz}{1 + z^2} \int_{\mathbb{R}} \left(\frac{1}{1 + x^2} + \frac{1}{1 + (z - x)^2} \right) dx = 2. \end{aligned}$$

The fractional moment in (13) follows from independence and the finite Cauchy moments of orders below one.

Here the stable limit has a short direct proof. For all real b, x ,

$$|\cos(b\sqrt{1 + x^2}) - \cos(bx)| \leq b^2/2, \quad (14)$$

since $y \mapsto \cos \sqrt{y}$ is $1/2$ -Lipschitz on $[0, \infty)$. Symmetry of W , (14), and the Cauchy characteristic function imply, for fixed (t, s) ,

$$\mathbb{E}e^{i(t\xi^{(1)} + s\xi^{(2)})/m} = \mathbb{E}e^{-|tW + s|/m} + O(m^{-2}) = 1 - \frac{\psi(t, s)}{m} + O(m^{-2}), \quad \psi(t, s) = \mathbb{E}|tW + s|. \quad (15)$$

The last remainder uses $\mathbb{E}W^2 < \infty$. Taking m th powers proves convergence and, by Lévy's continuity theorem, existence of (A, B) . In particular there is no centering term.

The symmetric, nonconstant W makes ψ continuous, homogeneous and positive away from zero. Hence $\psi(t, s) \geq c\sqrt{t^2 + s^2}$ for some $c > 0$, and Fourier inversion gives a continuous density. The support is convex: for independent copies Z_1, Z_2 of (A, B) , homogeneity gives $aZ_1 + (1 - a)Z_2 \stackrel{d}{=} (A, B)$ for $0 < a < 1$. Every nonzero linear projection is a nondegenerate Cauchy variable. A proper closed convex support would lie in a half-space, contradicting that projection property. Thus the support is all of $\mathbb{R}^2$.

Lines have probability zero, so A, B are nonzero almost surely and R has no atoms. Continuous mapping proves $R_m \Rightarrow R$; full planar support gives full support of R . Together with (12), this proves the bounds on V and its positive variance. $\square$

4 The joint limit, not a conditional-independence assumption

Let

$$\mathbb{G}_m(r) = \frac{1}{\sqrt{m}} \sum_{i=1}^m h_i(r), \quad r \neq 0.$$

We prove the process limit, its moments, and its independence from the stable vector in one step.

On either half-line, $p + rq$ changes sign at most once. At such a crossing the effective signs are mixed, so the jump of g has magnitude one. Thus, for $\delta \in \{-1, 1\}$,

$$h_i(\delta\rho) = \sigma_i(b_i^\delta + j_i^\delta \mathbf{1}_{\{\tau_i^\delta \leq \rho\}}), \quad \rho > 0, \quad (16)$$

where $|b_i^\delta|, |j_i^\delta| \leq 1$, $j_i^\delta \in \{-1, 0, 1\}$, and $\tau_i^\delta \in (0, \infty]$. All three coefficients depend only on the cosecants. A finite threshold has no atoms: condition on c_1, c_2 and use the continuous law of c_3 .

Functional convergence. Split the jump mark $J_i^\delta = \sigma_i j_i^\delta$ into its $+1$ and -1 cases, assigning threshold ∞ otherwise. Denote their empirical CDFs by $F_m^{\delta,\pm}$ and their CDFs by $F^{\delta,\pm}$. Centering (16) yields

$$\mathbb{G}_m(\delta\rho) = B_m^\delta + \sqrt{m}(F_m^{\delta,+} - F^{\delta,+})(\rho) - \sqrt{m}(F_m^{\delta,-} - F^{\delta,-})(\rho), \quad B_m^\delta = m^{-1/2} \sum_i \sigma_i b_i^\delta.$$

The empirical-CDF functional CLT [3, Theorem 1] applies on every compact subinterval of $(0, \infty)$. Indeed, quantile time change reduces each process to the uniform empirical process; the CDFs are continuous there, and an atom at ∞ is harmless after the map $x \mapsto x/(1+x)$. Marginal tightness of these four CDF processes and two baselines gives joint tightness. The multivariate CLT identifies all finite-dimensional limits, without assuming independence within a module. Hence $\mathbb{G}_m \Rightarrow \mathbb{G}$ locally in the Skorokhod topology on the two half-lines, with continuous Gaussian limiting paths and

$$\mathbb{E}\mathbb{G}(r)\mathbb{G}(s) = \mathbb{E}h(r)h(s), \quad \mathbb{E}\mathbb{G}(r)^2 = v(r). \quad (17)$$

Uniform moments. Condition on all cosecants. The σ_i are independent fair signs. For deterministic $|d_i| \leq 1$, the exponential-supermartingale bound gives

$$\mathbb{P} \left(\max_{k \leq m} \left| \sum_{i \leq k} \sigma_i d_i \right| > z\sqrt{m} \right) \leq 2e^{-z^2/2}.$$

This follows from $\cosh(td_i) \leq e^{t^2/2}$, the maximal inequality, and optimization at $t = z/\sqrt{m}$. In (16), order the jumps by threshold. The supremum is bounded by the absolute baseline sum plus the largest absolute partial jump sum. Apply the displayed bound to each, then use a union bound over the two half-lines:

$$\mathbb{P} \left(\sup_{r \neq 0} |\mathbb{G}_m(r)| > x \right) \leq 8e^{-x^2/8}. \quad (18)$$

Thus every fixed moment of this supremum is uniformly bounded. Right continuity makes the supremum measurable by restriction to rational $\rho > 0$.

Independence from the stable limit. Take any bounded centered linear combination $H = \sum_j a_j h(r_j)$ on one *original* module, and any $\zeta \in \mathbb{R}^2$. The fractional moment in (13) gives

$$\mathbb{E}|e^{i\zeta\xi/m} - 1| = O(m^{-3/4}).$$

Expand only $e^{iH/\sqrt{m}}$ to second order. The mixed linear term is bounded by $\|H\|_\infty m^{-1/2} O(m^{-3/4}) = o(m^{-1})$; the weighted quadratic term tends to $\mathbb{E}H^2$ by dominated convergence. Using (15) therefore gives

$$\mathbb{E}e^{i\zeta\xi/m + iH/\sqrt{m}} = 1 - \frac{\psi(\zeta) + \mathbb{E}H^2/2}{m} + o(m^{-1}).$$

Taking m th powers yields the product $e^{-\psi(\zeta)}e^{-\mathbb{E}H^2/2}$. Combined with the preceding tightness, this proves

$$(\mathbb{G}_m, U_m/m, T_m/m) \Rightarrow (\mathbb{G}, A, B), \quad \mathbb{G} \text{ independent of } (A, B). \quad (19)$$

Continuous paths and a countable dense set of evaluation points identify the process law and its independence. The marginal representation (13) has not replaced the actual joint law of h and ξ .

Since R is finite and nonzero, localize to $L^{-1} \leq |R_m| \leq L$, apply continuous mapping to (19), and let $L \rightarrow \infty$. This gives

$$\mathbb{G}_m(R_m) \Rightarrow \mathbb{G}(R) \stackrel{d}{=} Z\sqrt{v(R)}, \quad (20)$$

because, conditionally on R , the independent process has centered Gaussian value of variance $v(R)$.

5 Completion of the proof

Equation (9) gives

$$\tilde{X}_m := \frac{\tilde{S}_m - \beta_m/2}{\sqrt{m}} = \mathbb{G}_m(R_m) - \frac{\text{sgn } T_m}{2\sqrt{m}} \implies Z\sqrt{V}.$$

By (18), the $\tilde{X}_m$ have uniformly bounded moments of every fixed order. Uniform integrability therefore gives $\mathbb{E}\tilde{X}_m^2 \rightarrow \mathbb{E}V$ and $\mathbb{E}\tilde{X}_m^4 \rightarrow 3\mathbb{E}V^2$.

For the actual lengths (1), let ω_m be the total core-length fraction. Then

$$\omega_m \leq \frac{3}{m^5 + 3}, \quad \mathcal{L}(S_m) = (1 - \omega_m)\mathcal{L}(\tilde{S}_m) + \omega_m\mathcal{R}_m$$

by (6), with $\mathcal{R}_m$ a probability law. Thus the total-variation error is at most ω_m . All laws have center $\beta_m/2$ and support $[0, \beta_m]$, so for $p = 2, 4$,

$$\left| \mathbb{E}|X_m|^p - \mathbb{E}|\tilde{X}_m|^p \right| \leq \omega_m \left(\frac{\beta_m}{2\sqrt{m}} \right)^p = O(m^{p/2-5}) \rightarrow 0.$$

This proves (4); symmetry supplies the exact mean. Slutsky's theorem gives (5). Finally, for every $t \neq 0$, strict Jensen inequality gives

$$\mathbb{E}e^{-t^2V/(2a_*)} > e^{-t^2/2},$$

since V is nonconstant. The standardized limit is therefore not Gaussian, and its limiting standardized fourth moment is $3\mathbb{E}V^2/a_*^2 > 3$. $\square$

Other lengths and scope. On the same graphs the auxiliary law is length-independent. Hence any positive rationally independent lengths give the same weak limit on the $\sqrt{m}$ scale if $\omega_m \rightarrow 0$; the variance and standardized weak limits also hold if $m\omega_m \rightarrow 0$, and the fourth-moment limit holds if $m^2\omega_m \rightarrow 0$. These follow from the same bounds, not from total variation alone. The construction uses growing hub degrees and a dominant pendant length. No assertion is made for bounded degrees, comparable lengths, or joint spectral-window/graph-size limits.

References

- [1] L. Alon, R. Band and G. Berkolaiko, *Nodal statistics on quantum graphs*, Commun. Math. Phys. **362** (2018), 909–948. arXiv:1709.10413v2, Theorem 2.1.
- [2] L. Alon, R. Band and G. Berkolaiko, *Universality of nodal count distribution in large metric graphs*, Experimental Mathematics **33** (2024), doi:10.1080/10586458.2022.2092565. All locators refer to arXiv:2106.06096v2: Conjecture 3.1, Proposition 6.1, Eq. (6.7), and Lemma 6.7.
- [3] J.-F. Marckert, *One more approach to the convergence of the empirical process to the Brownian bridge*, Electron. J. Stat. **2** (2008), 118–126. arXiv:0710.3296v2, Theorem 1.